

Local Setup Guide

AI Business Card CRM — MSTinternship Project

Step 1 — Prerequisites

Install all of the following before proceeding:

Tool	Purpose	Where to Get It
Python 3.11+	Runs Django	python.org
PostgreSQL	Database	postgresql.org
Redis	Background task queue	redis.io
Ollama	Runs AI models locally	ollama.com
Git	Clone the repository	git-scm.com

Step 2 — Clone the Repository

```
git clone https://github.com/Rehansh26/MSTinternship-Rehansh-Shrivastava.git
cd MSTinternship-Rehansh-Shrivastava
```

Step 3 — Create a Virtual Environment

Mac / Linux:

```
python3 -m venv venv
source venv/bin/activate
```

Windows:

```
python -m venv venv
venv\Scripts\activate
```

Step 4 — Install Dependencies

```
pip install -r requirements.txt
```

Step 5 — Set Up the Database (PostgreSQL)

Open pgAdmin or psql and run:

```
CREATE DATABASE crm_db;
```

Step 6 — Create the .env File

Create a file named **.env** in the project root folder with the following contents:

```
SECRET_KEY=any-long-random-string-here
DB_NAME=crm_db
DB_USER=postgres
DB_PASSWORD=your_postgres_password
DB_HOST=localhost
DB_PORT=5432
```

Step 7 — Pull the AI Models (one-time, large downloads)

```
ollama pull llama3.2-vision # reads business card images (~8 GB)
ollama pull llama3.2        # chat assistant (~2 GB)
ollama pull nomic-embed-text # semantic search (~300 MB)
```

Step 8 — Run Database Migrations

```
python manage.py migrate
python manage.py createsuperuser
```

The `createsuperuser` command sets your login username and password for the app.

Step 9 — Start All 4 Services

Open 4 separate terminal tabs, all inside the project folder with the `venv` activated:

Tab 1 — Django web server:

```
python manage.py runserver
```

Tab 2 — Celery worker (processes card scans in background):

```
celery -A card_manager worker --loglevel=info
```

Tab 3 — Redis (if not running as a system service):

```
redis-server
```

Tab 4 — Ollama (if not running as a system service):

```
ollama serve
```

Step 10 — Open the App

Go to **`http://127.0.0.1:8000`** in your browser and log in with the superuser credentials you created in Step 8.

Quick Checklist

- PostgreSQL running and `crm_db` database exists
- `.env` file created with correct DB password
- `pip install -r requirements.txt` completed with no errors
- All 3 Ollama models downloaded
- `python manage.py migrate` ran successfully
- All 4 services running (Django, Celery, Redis, Ollama)

Common Errors

Error	Fix
connection refused on DB	PostgreSQL is not running, or wrong password in <code>.env</code>
connection refused on Redis	Start Redis with: <code>redis-server</code>
Card stuck on "AI is scanning"	Celery is not running, or Ollama is not running
<code>ModuleNotFoundError</code>	Venv not activated, or <code>pip install</code> not done